

Math 521, Section 2
Homework 12
Selected Solutions

Problem 1. Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \rightarrow Y$, and $x_0 \in X$. Prove that f is continuous at x_0 if and only if for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$.

Solution: $\Rightarrow$ Suppose that f is continuous at x_0 . Fix an open set $V \subseteq Y$ so that $f(x_0) \in V$. Then there exists $\epsilon > 0$ so that $B_\epsilon^Y(f(x_0)) \subseteq V$. As f is continuous at x_0 , then there exists $\delta > 0$ so that

$$d_X(x, x_0) < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x_0)) < \epsilon.$$

Set $U = B_\delta^X(x_0)$. Then U is an open subset of X that contains x_0 . Moreover, $x \in U$ implies $f(x) \in V$, since

$$\begin{aligned} x \in U = B_\delta^X(x_0) &\Longrightarrow d_X(x, x_0) < \delta \\ &\Longrightarrow d_Y(f(x), f(x_0)) < \epsilon \\ &\Longrightarrow f(x) \in B_\epsilon^Y(f(x_0)) \subseteq V. \end{aligned}$$

$\Leftarrow$ Assume that for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$. We want to show that f is continuous at x_0 . Fix $\epsilon > 0$. Then $V = B_\epsilon^Y(f(x_0))$ is an open subset of Y containing $f(x_0)$. So, we know that there exists an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$. As U is open, there must exist $\delta > 0$ so that $B_\delta^X(x_0) \subseteq U$. Altogether, we conclude

$$\begin{aligned} d_X(x, x_0) < \delta &\Longrightarrow x \in B_\delta^X(x_0) \subseteq U \\ &\Longrightarrow f(x) \in V = B_\epsilon^Y(f(x_0)) \\ &\Longrightarrow d_Y(f(x), f(x_0)) < \epsilon \end{aligned}$$

as desired. □

Problem 2. Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \rightarrow Y$ a function. Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

Solution: $\boxed{\implies}$ Suppose that f is continuous. Fix $A \subseteq X$ and $y \in f(\overline{A})$; we need to show that $y \in \overline{f(A)}$ as well. As $y \in f(\overline{A})$, then there exists $x_0 \in \overline{A}$ so that $f(x_0) = y$. Then, there exists a sequence $\{x_n\}$ in A so that $x_n \rightarrow x_0$ (by Lecture 25). As f is continuous at x_0 , then $f(x_n) \rightarrow f(x_0)$ (by Lecture 30). So $\{f(x_n)\}$ is a sequence in $f(A)$ that converges to $f(x_0)$, and so $f(x_0) \in \overline{f(A)}$.

$\boxed{\impliedby}$ Assume that $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$. Suppose towards a contradiction that f were not continuous. Then there would exist a sequence $\{x_n\}$ in X so that x_n converges to x_0 but $f(x_n)$ does not converges to $f(x_0)$. Negating the definition of ‘ $f(x_n)$ converges to $f(x_0)$ ’, we get: there exists $\epsilon > 0$ so that for any N , there is some $n \geq N$ so that $d_Y(f(x_n), f(x_0)) \geq \epsilon$.

- Taking $N = 1$, there must exist some $k_1 \geq 1$ so that $d_Y(f(x_{k_1}), f(x_0)) \geq \epsilon$.
- Taking $N = k_1 + 1$, there must exist some $k_2 > k_1$ so that $d_Y(f(x_{k_2}), f(x_0)) \geq \epsilon$.

Proceeding inductively, we get a subsequence $\{x_{k_n}\}$ so that $d_Y(f(x_{k_n}), f(x_0)) \geq \epsilon$ for all $n \geq 1$.

Consider the set $A = \{x_{k_n} : n \in \mathbb{N}\}$, and note that $x_0 \in \overline{A}$ since $x_{k_n} \rightarrow x_0$. For this set we must have $f(\overline{A}) \subseteq \overline{f(A)}$, and so $f(x_0) \in \overline{f(A)} \subseteq \overline{f(A)}$. This implies that $B_\epsilon(f(x_0)) \cap f(A) \neq \emptyset$, and so there exists a point $f(x_{k_n}) \in f(A)$ so that $d_Y(f(x_{k_n}), f(x_0)) < \epsilon$. However, this contradicts how x_{k_n} was chosen. $\square$

Problem 4. Consider the metric spaces $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^2, d_2)$.

- (a) Prove that the functions $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = x$ and $g(x, y) = y$ are continuous. (Hint: Use sequences.)
- (b) Prove that if $h : [0, 1] \rightarrow \mathbb{R}^2$ is a continuous function such that $h(0) = (x_0, y_0)$, $h(1) = (x_1, y_1)$, and $x_0 < 0 < x_1$, then there exists a point $t \in (0, 1)$ so that $h(t)$ lies on the y -axis.

Solution: (a) Fix a sequence (x_n, y_n) in $(\mathbb{R}^2, d_2)$ that converges to some point (x_0, y_0) . By Lecture 30, it suffices to show that $f(x_n, y_n)$ converges to $f(x_0, y_0)$ and $g(x_n, y_n)$ converges to $g(x_0, y_0)$ in $(\mathbb{R}, |\cdot|)$. Notice that $f(x_n, y_n) = x_n$ and $g(x_n, y_n) = y_n$. As $(x_n, y_n) \rightarrow (x_0, y_0)$ in $(\mathbb{R}^2, d_2)$, then by HW9#5 we know that both $x_n \rightarrow x_0$ and $y_n \rightarrow y_0$ in $(\mathbb{R}, |\cdot|)$.

(b) Write $h(t) = (x(t), y(t)) \in \mathbb{R}^2$. It suffices to show that there exists a point $t \in (0, 1)$ so that $x(t) = 0$. We will argue using the Intermediate Value Theorem. Note that $x : [0, 1] \rightarrow \mathbb{R}$ satisfies $x(0) = x_0 < 0 < x_1 = x(1)$. So it only remains to show that $x(t)$ is continuous.

Claim: $x : [0, 1] \rightarrow \mathbb{R}$ is a continuous function. Note that $x = f \circ h$, where f is the continuous function from part (a). So x is the composition of two continuous functions, and thus is also continuous. $\square$

Problem 5. Prove that the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum.

Solution: We know that the set $[-1, 1]^2$ is compact in $\mathbb{R}^2$ by Lecture 28. Moreover, the function f is continuous on $[-1, 1]^2$ since it is the sum and composition of polynomials and $\sin$. So, by the extreme value theorem, there must exist a point $(x_0, y_0) \in [-1, 1]^2$ so that

$$f(x, y) \geq f(x_0, y_0) \quad \text{for all } (x, y) \in [-1, 1]^2.$$

For $|x| > 1$, note that

$$f(x, y) = x^2 + y^2 + \sin(x - y) \geq 1 + 0 - 1 = 0.$$

Similarly, when $|y| > 1$ we have

$$f(x, y) = x^2 + y^2 + \sin(x - y) \geq 0 + 1 - 1 = 0.$$

Together, this yields

$$f(x, y) \geq 0 \quad \text{for all } (x, y) \in \mathbb{R}^2 \setminus [-1, 1]^2.$$

On the other hand, notice that $f(0, 0) = 0$. As $(0, 0)$ is a point in $[-1, 1]^2$, then the minimum of f on $[-1, 1]^2$ must be smaller than this:

$$0 = f(0, 0) \geq f(x_0, y_0).$$

Combining this with the previous inequality, we see that

$$f(x, y) \geq f(x_0, y_0) \quad \text{for all } (x, y) \in \mathbb{R}^2 \setminus [-1, 1]^2.$$

Collecting the previous two paragraphs, we conclude that

$$f(x, y) \geq f(x_0, y_0) \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Therefore (x_0, y_0) is the global minimum of f . □

Problem 6. Let X and Y be metric spaces and $f : X \rightarrow Y$ a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \rightarrow X$ is also continuous.

Solution: It suffices to prove that for any closed set $F \subseteq X$, the set $(f^{-1})^{-1}(F)$ is closed in Y . Fix $F \subseteq X$ closed. As F is a closed subset of the compact space X , then F is compact as well (by HW10#3). Then $f(F)$ is also compact (by Lecture 32), since f is continuous. In particular, $f(F)$ is a closed subset of Y (by Lecture 27).

Finally, note that the set $f(F)$ is the same as the set $(f^{-1})^{-1}(F)$. Indeed, in the special case where g is a bijective function, then the preimage $g^{-1}(F)$ of F under g is the same thing as the image of F under the inverse function g^{-1} . Here, g is the bijective function f^{-1} , and the inverse function g^{-1} is simply f , since $(f^{-1} \circ f)(x) = x$ for all $x \in X$ and $(f \circ f^{-1})(y) = y$ for all $y \in Y$. $\square$